
SPECIFICATION FOR COSPAS-SARSAT 406 MHz DISTRESS BEACONS

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SPECIFICATION FOR COSPAS-SARSAT
406 MHz DISTRESS BEACONS

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1. INTRODUCTION

1.1 Purpose

The purpose of this document is to define the minimum requirements to be used for the development and manufacture of 406 MHz Emergency Locator Transmitters (ELTs), Emergency Position-Indicating Radio Beacons (EPIRBs), and Personal Locator Beacons (PLBs). In this document, the term ELT indicates an aviation distress beacon, an EPIRB a maritime distress beacon, and a PLB a distress beacon for personal use.

Specifications that are critical to the Cospas-Sarsat System are defined in detail; specifications which could be developed by the national authorities are identified in more general terms.

Where there are specific requirements in this document for a distress tracking ELT (ELT(DT)) these refer to a specific type of ELT designed to be activated prior to a crash and to function in compliance with the ICAO GADSS* requirements for the location of an aeroplane in distress, aimed at establishing, to a reasonable extent, the location of an accident site within a 6 NM radius. An ELT(DT) may be activated automatically upon detection of a distress condition while in flight or it may also be activated manually.

1.2 Scope

This document contains the minimum requirements that apply to Cospas-Sarsat 406 MHz distress beacons. It is divided into the following sections:

- a) Section 2 gives the system requirements applicable to all types of beacons. When met, these requirements will enable the beacons to provide the intended service in terms of location probability and accuracy and will not disturb the system operation.
- b) Section 3 deals with the beacon message content. Basic message structure is defined. Assignment and meaning of the available data bits are defined in Annex A to this specification.
- c) Section 4 defines a set of environmental and operational requirements. These requirements are not intended to be exhaustive and may be complemented by more detailed national or international standards (e.g., RTCA standards for ELTs). However, they represent the minimum environmental and operational performance requirements for a 406 MHz beacon to be compatible with the Cospas-Sarsat System.

* See ICAO Convention on International Civil Aviation Annex 6, Part 1.

- d) Annex A defines the beacon coding.
- e) Annex B provides samples of error correcting code calculations, M_{offset} calculations and CRC code sample.

A full list of acronyms used in this document can be found in document C/S S.011, “Cospas-Sarsat Glossary”.

- END OF SECTION 1 -

2. SYSTEM REQUIREMENTS

2.1 Beacon Functional Elements

This section defines the requirements for the two following functional elements of a 406 MHz distress beacon:

- a) digital message generator; and
- b) modulator and 406 MHz transmitter.

2.2 Digital Message Generator

The digital message generator will key the modulator and transmitter so that the message defined in section 3 is transmitted.

2.2.1 Repetition Period

The repetition period shall not be so stable that any two transmitters appear to be synchronized closer than a few seconds over a 5-minute period. The intent is that no two beacons will have all of their bursts coincident. The period shall be randomised around a mean value of 50 seconds, so that time intervals between transmissions are randomly distributed on the interval 47.5 to 52.5 seconds.

For ELT(DT)s (except those transmitting the 3LD* aircraft operator designator in PDF-2) the value of the repetition period shall be as follows:

- a) From beacon activation, transmit a total of 24 bursts, the time interval between the beginning of two successive bursts shall be 5 seconds ± 0.0 / - 0.2 seconds. The first burst timing shall meet the requirements of 4.5.6;
- b) Then transmit a further 18 bursts commencing 10 seconds ± 0.0 / - 0.2 after the burst 24, at time intervals between the beginning of two successive bursts of 10 seconds ± 0.0 / - 0.2 seconds; and
- c) The next burst (burst number 43) shall then start between 27.0 and 30.0 seconds after the beginning of burst 42. Subsequent bursts shall be randomized with uniform (flat) distribution around a mean value of 28.5 seconds so that the time intervals between the beginnings of two successive bursts (T_R) are randomly distributed on the interval 27.0 to 30.0 seconds. The standard deviation over

* 3LD is a 3-letter aircraft operator designator from the list of "Designators for Aircraft Operating Agencies, Aeronautical Authorities and Services" published by the International Civil Aviation Organization (ICAO) as document 8585.

18 successive bursts of T_R shall be greater than 0.8 seconds. The minimum value of T_R observed over the 18 successive bursts shall be between 27.0 and 27.2 seconds, the maximum value of T_R observed over the 18 successive bursts shall be between 29.8 and 30.0 seconds.

For ELT(DT)s transmitting the 3LD aircraft operator designator in PDF-2, the transmission scheme is as follows. An illustrative version of this is provided in Figure 2.1.

From beacon activation, transmit a total of 24 bursts, the time interval between the beginning of two successive bursts shall be 5 seconds $+0.0 / - 0.2$ seconds and every 4th burst starting with burst 2 shall transmit the 3LD.

The first burst timing shall meet the requirements of 4.5.6.

Then transmit a further 18 bursts commencing 10 seconds $+ 0.0 / - 0.2$ after the 24th burst, at time intervals between the beginning of two successive bursts of 10 seconds $+ 0.0 / - 0.2$ seconds. In addition, transmit an additional burst, containing the 3LD, 5 seconds ± 0.5 seconds after the beginning of the first of these 18 bursts and then after every 6th burst (i.e. 3 additional bursts with the 3LD, interleaved between the 18 bursts, making a total of 21 bursts).

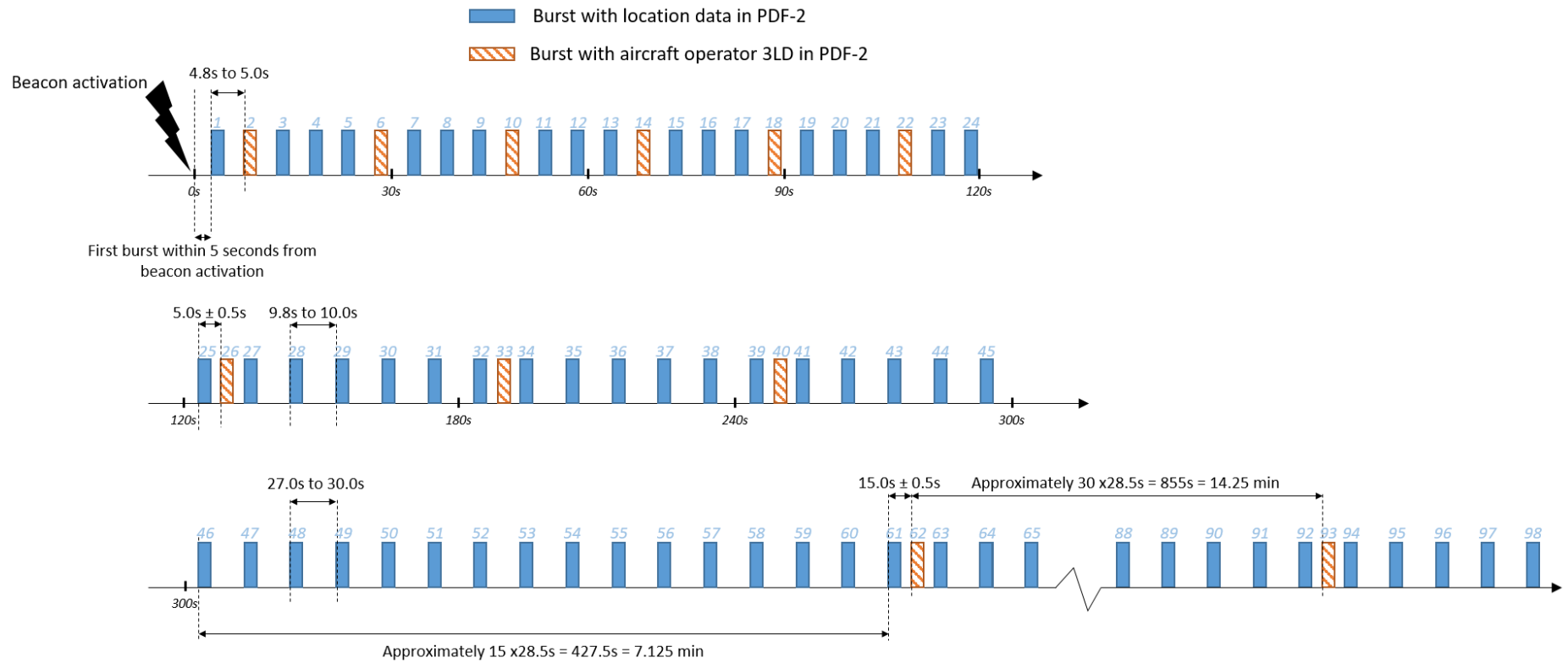
The next burst shall then start between 27.0 and 30.0 seconds after the beginning of the last (45th) burst and subsequent bursts shall be randomized with uniform (flat) distribution around a mean value of 28.5 seconds so that the time intervals between the beginnings of two successive bursts (T_R) are randomly distributed on the interval 27.0 to 30.0 seconds. In addition, an additional burst containing the 3LD shall be transmitted after the 16th of these bursts and then after every 30th burst thereafter at 15 seconds ± 0.5 seconds after the beginning of the previous burst. The standard deviation over 18 successive bursts of T_R (excluding bursts containing the 3LD) shall be greater than 0.8 seconds. The minimum value of T_R observed over the 18 successive bursts shall be between 27.0 and 27.2 seconds, and the maximum value of T_R observed over the 18 successive bursts shall be between 29.8 and 30.0 seconds (ignoring bursts containing the 3LD).

2.2.2 Total Transmission Time

The total transmission time, measured at the 90 percent power points, shall be 440 ms ± 1 percent for the short message and 520 ms ± 1 percent for the long message.

2.2.3 Unmodulated Carrier

The initial 160 ms ± 1 percent of the transmitted signal shall consist of an unmodulated carrier at the transmitter frequency measured between the 90 percent power point and the beginning of the modulation.



Note - The times in seconds under the axis of each set of bursts are for reference only and do not necessarily indicate where bursts will occur as timing tolerances will vary this.

Figure 2.1: Burst Transmission Scheme for ELT(DT)s transmitting the 3LD aircraft operator designator in PDF-2

2.2.4 Digital Message

a) Short Message

The final 280 ms ± 1 percent of the transmitted signal shall contain a 112-bit message at a bit rate of 400 bps ± 1 percent;

b) Long Message

The final 360 ms ± 1 percent of the transmitted signal shall contain a 144-bit message at a bit rate of 400 bps ± 1 percent.

For ELT(DT)s, and RLS location protocol beacons, the final 360 ms ± 1 percent of the transmitted signal shall contain a 144-bit message at a bit rate of 400 bps ± 0.1 percent.

c) Cancellation Message (ELT(DT) only)

ELT(DT)s shall also have means of deactivation by the same means of activation which shall be followed by a cancellation message sequence, as defined in section 3.3.

2.2.4.1 Bit Synchronization

A bit-synchronization pattern consisting of "1"s shall occupy the first 15-bit positions.

2.2.4.2 Frame Synchronization

A frame synchronization pattern consisting of 9 bits shall occupy bit positions 16 through 24. The frame synchronization pattern in normal operation shall be 000101111. However, if the beacon radiates a modulated signal in the self-test mode, the frame synchronization pattern shall be 011010000 (i.e. the last 8 bits are complemented).

2.2.4.3 Format Flag

Bit 25 is a format (F) flag bit used to indicate the length of the message to follow. Value "0" indicates a short message; value "1" indicates a long message.

2.2.4.4 Message Content

The content of the remaining 87 bits (short message - see Figure 2.2) or 119 bits (long message - see Figure 2.3) is defined in section 3.

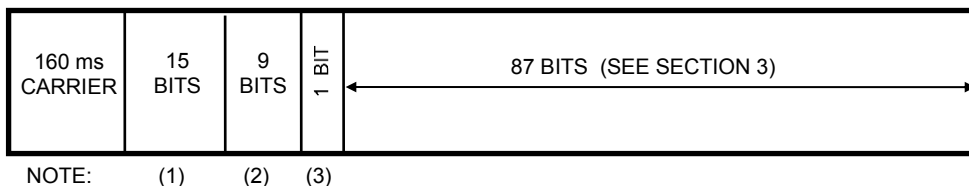


Figure 2.2: Short Message Format

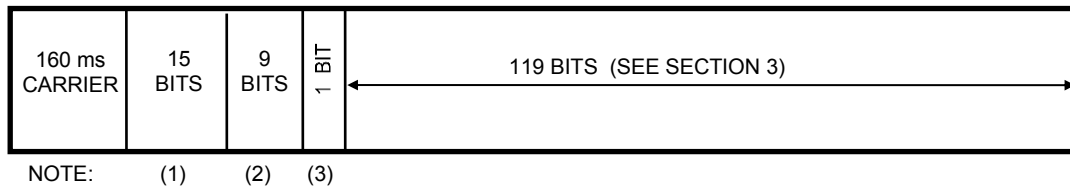


Figure 2.3: Long Message Format

- Notes:
- (1) Bit Synchronization : 15 "1" bits
 - (2) Frame Synchronization : 000101111 (except as in section 4.5.4)
 - (3) "0" bit indicates short-message format
"1" bit indicates long-message format

2.3 Modulator and 406 MHz Transmitter

2.3.1 Transmitted Frequency*

To ensure adequate System capacity and an efficient use of the available frequency spectrum in the band 406.0 - 406.1 MHz allocated by the ITU for the operation of low-power satellite emergency position-indicating radiobeacons, a number of channels have been defined in the allocated band and will be assigned by Cospas-Sarsat from time to time, as necessary to satisfy capacity requirements.

The frequency channels in the band 406.0 - 406.1 MHz are defined by the centre frequency of the channels, as assigned by Cospas-Sarsat.

Except as provided below for beacons type approved by Cospas-Sarsat for operation at 406.025 MHz and 406.028 MHz, the beacon carrier frequency shall be set in accordance with the Cospas-Sarsat 406 MHz Channel Assignment Table, as provided in document C/S T.012 "Cospas-Sarsat 406 MHz Frequency Management Plan", at the designated centre frequency of the appropriate channel ± 1 kHz, and shall not vary more than ± 5 kHz from that channel centre frequency in 5 years.

The carrier frequency of beacons operating in the 406.025 MHz channel in accordance with the Cospas-Sarsat 406 MHz Channel Assignment Table shall be set at 406.025 MHz ± 2 kHz. The carrier frequency shall not vary more than ± 5 kHz from 406.025 MHz in 5 years.

The carrier frequency of beacons operating in the 406.028 MHz channel in accordance with the Cospas-Sarsat 406 MHz Channel Assignment Table shall be set at 406.028 ± 1 kHz. The carrier frequency shall not vary more than $+2$ kHz / -5 kHz from 406.028 MHz in 5 years.

* This section of the beacon specification does not apply to Cospas-Sarsat System beacons (i.e., orbitography or calibration beacons). The transmitted frequency requirements for orbitography beacons are detailed in document C/S T.006.

The transmitted frequency short-term variations shall not exceed 2 parts in 10^9 in 100 ms.

The transmitted frequency medium-term stability shall be defined by the mean slope of the frequency versus time over a 15-minute period and by the residual frequency variation about the mean slope. The mean slope shall not exceed 1 part in 10^9 per minute, except as noted below. The residual frequency variation shall not exceed 3 parts in 10^9 .

After allowing 15-minutes for beacon warm-up, the medium-term frequency stability requirements shall be met for all defined environmental conditions, except for the temperature gradient and the thermal shock as defined in sections 4.2.2 and 4.2.3 respectively.

The mean slope of the medium-term frequency stability measurements shall not exceed 2 parts in 10^9 per minute, and the residual frequency variation shall not exceed 3 parts in 10^9 :

- during the variable temperature conditions of the temperature gradient ($\pm 5^\circ \text{C/h}$ slope) defined in section 4.2.2 and for the 15 minute periods immediately after the temperature had stabilised at the maximum or minimum values; and
- during the thermal shock defined in section 4.2.3.

It is recommended that distress transmissions commence as soon as possible after activation, but in accordance with section 4.5.6.

The mean slope and the residual frequency variation shall be measured as follows: Data shall be obtained by making 18 sequential frequency measurements, one every repetition period ($50 \text{ sec} \pm 5 \text{ percent}$, see section 2.2.1) over an approximate 15-minute interval; each measurement shall be a 100-ms frequency average performed during the modulated part of the message.

The mean slope is defined as that of the least-squares straight-line fit to the 18 data points. Residual frequency variation is defined as the root mean square (RMS) error of the points relative to the least-squares estimate.

For ELT(DT)s, the requirement of medium-term stability (mean slope and residual frequency variation) does not apply.

2.3.2 Transmitter Power Output

For all beacon types other than ELT(DT), the transmitter power output shall be within the limits of 35 to 39 dBm measured into a 50-Ohm load. This power output shall be maintained during the minimum operating lifetime of 24 hours (or for a longer period, as specified by National Administrations, or by International Organisations, or as declared by the beacon manufacturer, if it is different from 24 hours), at any temperature throughout the specified operating temperature range. Power output rise time shall be less than 5 ms measured between the 10% and 90% power points. The power output is assumed to rise linearly from zero and therefore must be zero prior to about 0.6 ms before the beginning of the rise time measurement; if it is not zero, the maximum acceptable level is -10 dBm.

For ELT(DT)s, the transmitter power output shall be within the limits of 36 dBm to 39 dBm measured into a 50-Ohm load. This power output shall be maintained during the minimum operating lifetime of 370-minutes (or for a longer period, as specified by National Administrations or by International

Organizations, or as declared by the beacon manufacturer, if it is different from 370 minutes), or for an ELT(DT) specifically designed to withstand a crash, or an ELT(DT) combined with an Automatic ELT for the Duration of Continuous Operation as defined in either 4.5.10.5 or 4.5.11.7 as applicable, at any temperature throughout the specified operating temperature range. Power output rise time shall be less than 2 ms measured between the 10% and 90% power points. The power output is assumed to rise linearly from zero and therefore must be zero prior to about 0.6 ms before the beginning of the rise time measurement; if it is not zero, the maximum acceptable level is -10 dBm.

2.3.3 Antenna Characteristics

The following antenna characteristics are defined for all azimuth angles and for elevation angles greater than 5° and less than 60°:

- Pattern : hemispherical
- Polarization : circular (RHCP) or linear
- Gain : between:
 - 3 dBi and 4 dBi over 90% of the above region
 - 2 dBi and 6 dBi over 90% of the above region for ELT(DT)s
- Antenna VSWR : not greater than 1.5:1

The antenna characteristics should be measured in a configuration as close as possible to its operational condition.

2.3.4 Spurious Emissions

The in-band spurious emissions shall not exceed the levels specified by the signal mask in Figure 2.4, when measured in a 100 Hz resolution bandwidth.

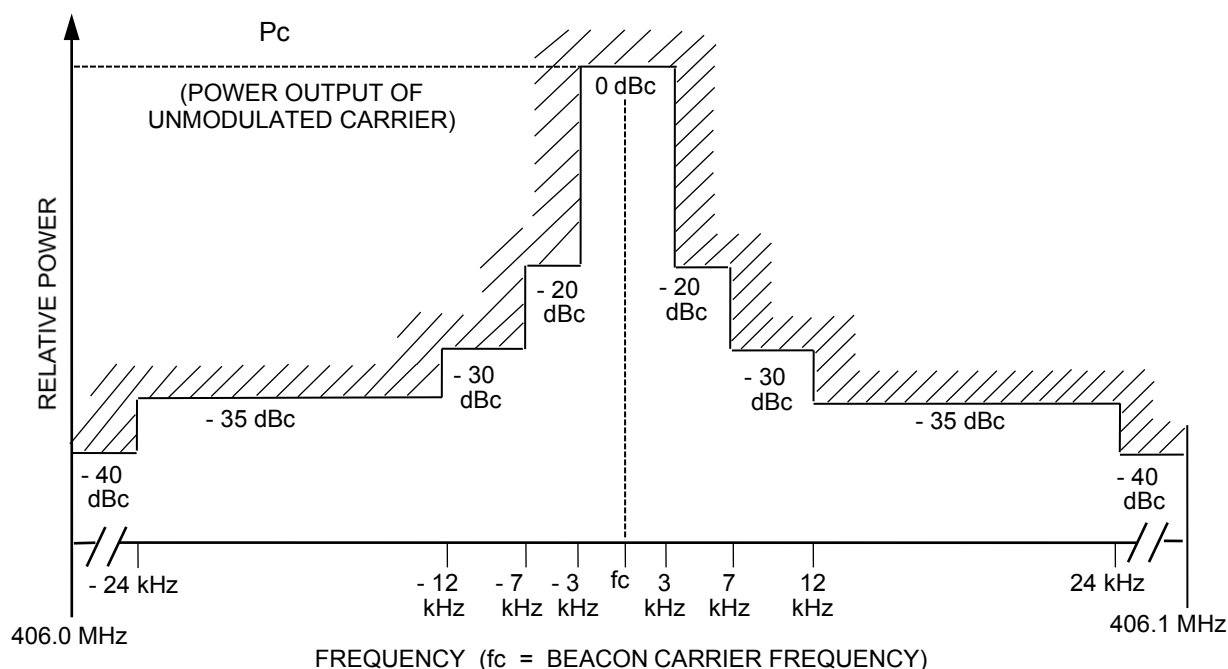


Figure 2.4: Spurious Emission Mask for 406.0 to 406.1 MHz Band

2.3.5 Data Encoding

The data shall be encoded biphas L, as shown in Figure 2.5.

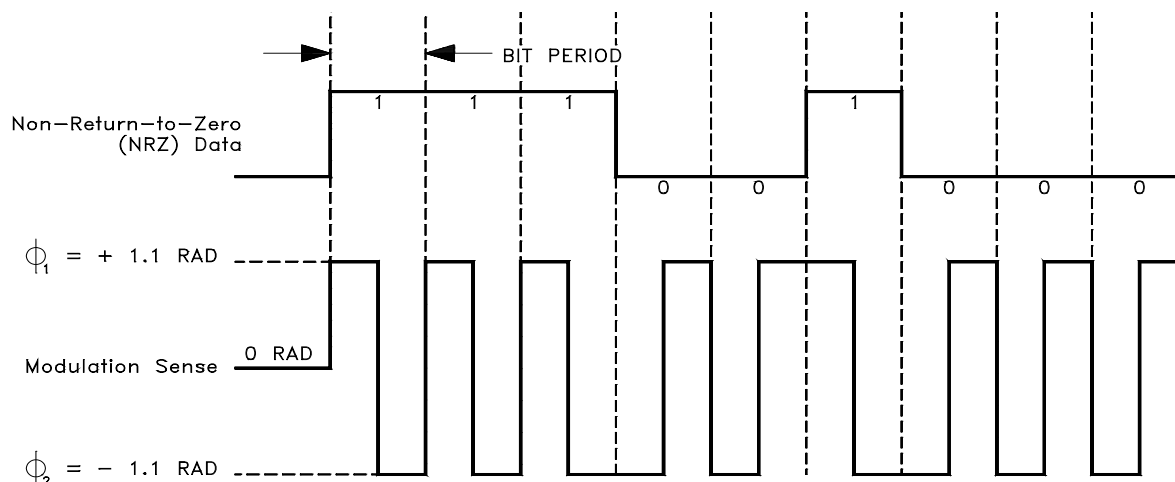


Figure 2.5: Data Encoding and Modulation Sense

2.3.6 Modulation

The carrier shall be phase modulated positive and negative 1.1 ± 0.1 radians peak, referenced to an unmodulated carrier. Positive phase shift refers to a phase advance relative to nominal phase. Modulation sense shall be as shown in Figure 2.5.

The rise (τ_R) and fall (τ_F) times of the modulated waveform, as shown in Figure 2.6, shall be $150 \pm 100 \mu\text{s}$.

For ELT(DT)s, the rise (τ_R) and fall (τ_F) times of the modulated waveform, shall be between $50 \mu\text{s}$ and $150 \mu\text{s}$.

$$\frac{|\tau_1 - \tau_2|}{\tau_1 + \tau_2} \leq 0.05$$

Modulation symmetry (see Figure 2.7) shall be such that:

2.3.7 Voltage Standing-Wave Ratio

The modulator and 406 MHz transmitter shall be able to meet all requirements, except for those in paragraph 2.3.2 (transmitter power output), at any VSWR between 1:1 and 3:1, and shall not be damaged by any load from open circuit to short circuit.

2.3.8 Maximum Continuous Transmission

The distress beacon shall be designed to limit any inadvertent continuous 406 MHz transmission to a maximum of 45 seconds.

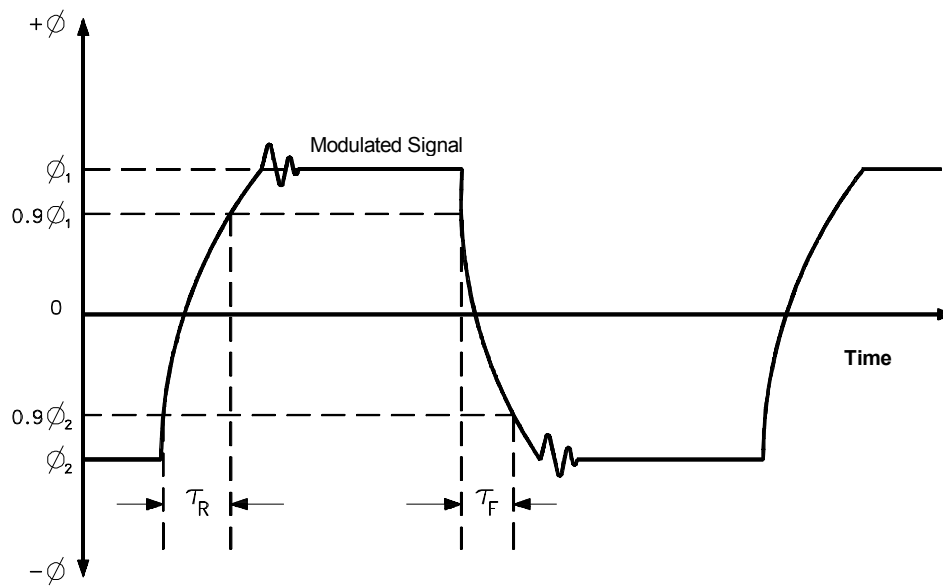


Figure 2.6*: Definition of Modulation Rise and Fall Times

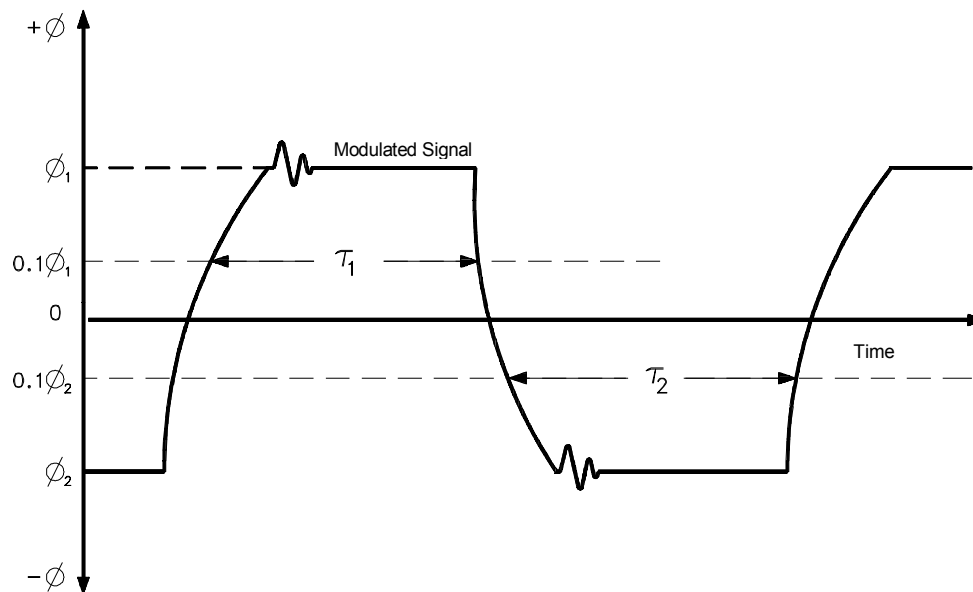


Figure 2.7†: Definition of Modulation Symmetry

- END OF SECTION 2 -

* Figure not to scale.

† Figure not to scale.